AIP Conference Proceedings

RESEARCH ARTICLE | JUNE 16 2026

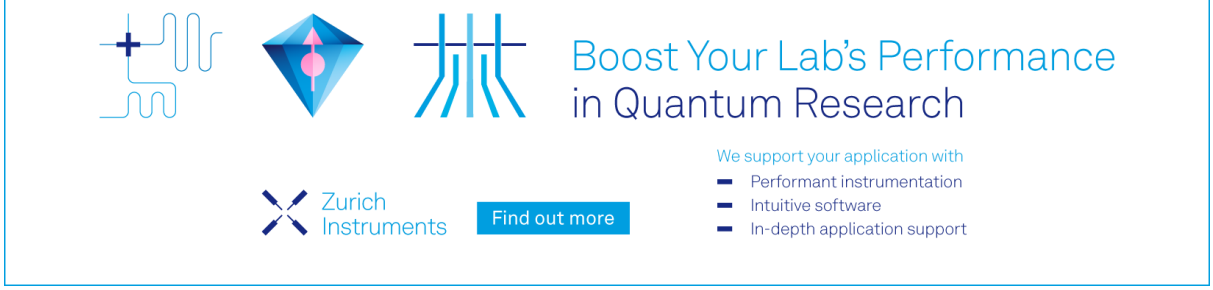
An efficient feature extraction method for emotion classification using peripheral signals

P. Y. Preema ✉; J. Chandra; Alwin Joseph; Joel John Joseph

AIP Conf. Proc. 3432, 050028 (2026)
https://doi.org/10.1063/5.0328667

View Online
Export Citation

09 October 2026 04:16:41



AIP Publishing

An Efficient Feature Extraction Method for Emotion Classification Using Peripheral Signals

P Y Preema[a], J Chandra[b], Alwin Joseph[c], and Joel John Joseph[d]

Christ University, Bangalore, India

[a)] Corresponding author: preema.py@res.christuniversity.in,
[b)] chandra.j@christuniversity.in
[c)] alwin.joseph@christuniversity.in,
[d)] joeljoseph1810@gmail.com

Abstract. Emotion Classification is a complex task impacted by several variables. The research aims to create a precise model for identifying and interpreting emotional states. The objective of the study is to identify peripheral signals that are more relevant to classifying human emotions accurately. This study proposes an approach to emotion classification by integrating six peripheral physiological signals Skin Temperature (ST), Respiration (RSP), Galvanic Skin Response (GSR), Electrooculography (EOG), Electromyography (EMG), and Photoplethysmography (PPG), The involuntary, and continuous nature of peripheral signals makes them especially promising for emotion categorization since they prevent users from manipulating or fabricating feelings for incorrect classification. The emotional states were identified using the amplitude domain signal components retrieved from the Fast Fourier Transform combined to form a fused feature set. Machine Learning (ML) Algorithms were used for the classification of emotions. Random Forest (RF), Extreme Gradient Boosting (XGBoost), and Light Gradient Boosting Machine (LGBM) showed the highest accuracy for classifying emotional states. XGBoost has been proven to perform better for all emotional states and outperform other machine learning models. The highest accuracy was for recognizing the High Arousal Low Valence (HALV), with an accuracy of 90.57%. The proposed methodology holds great promise for real-world applications, particularly in wearable technology and remote healthcare systems. Accurate identification of emotional states, especially negative emotions such as HALV, can aid in the early detection of stress, mental health issues, and even suicidal tendencies. In the future, more effective models can be designed to create a more responsive interface to enhance user experiences across various applications, from gaming to virtual assistants.

09 October 2026 04:16:41

INTRODUCTION

Emotional state identification has gained significant attention due to its profound implications in healthcare, mental well-being, and the prevention of adverse outcomes such as suicide. Many everyday decisions and day-to-day social interactions, through which people share their personal experiences, have a profound impact on one's welfare that is so deep and fundamental that it frequently has a profound nature. This research aims to bring to our attention a recent novel approach designed to consider emotions more sensitively within the framework of technologies, primarily for enhancing Human-Computer Interaction (HCI) in meaningful ways and making the interaction highly effective. The system analyzes peripheral signals such as Skin Temperature (ST), Respiration (RSP), Galvanic Skin Response (GSR), Electrooculography (EOG), Electromyography (EMG), and Photoplethysmography (PPG), and numerous studies have utilized peripheral signals for emotion recognition, the integration of these modalities remains underexplored. The study transcends traditional methods, examining peripheral signals to classify emotional states. The study utilized data from 32 participants and offered the most up-to-date ML mechanisms to process the deep-lying signals fully. Therefore, many such signals emanate from peripheral signs whose nature is essentially involuntary, deriving from the actual and physiological activity of the autonomic nervous system. Although many methods are applied to monitor this aspect, they may only produce results based on measures acquired from facial expression-based methods and the individual's voice recognition mechanism. This makes them extremely useful for a plethora of applications where precision and accuracy are not only highly desirable but also critical, particularly for Artificial Intelligence (AI) adaptive interfaces or assistive technology designed for individuals with disabilities.

Emotional awareness extends beyond technological progress and has significant implications for mental health and well-being. Envision the ability to wear something that may track your emotional state with little-to-no obtrusiveness in real-time, so patterns of anxiety or depression emerge before becoming uncontrollable conditions. This could be the essential trigger for such systems to activate early warning mechanisms that would assist

ETLTC2025 International Conference Series
AIP Conf. Proc. 3432, 050028-1–050028-14; https://doi.org/10.1063/5.0328667
Published under an exclusive license by AIP Publishing. 978-0-7354-5404-0/$30.00

professionals in mental health and care in intervening. It can also empower people to recognize their emotional pain and respond accordingly, leading to greater self-awareness and resilience. AI will also help enrich the empathy present in virtual communication, thereby enhancing the overall experience of interaction. For instance, virtual assistants can notice and pick up on undertones in the user's voice that might reveal whether the user is stressed or if they are not in a good mood about something. Because of this, these assistants will adjust and modify their communication pattern and approach based on the need to address the stress the user is experiencing. Gradually, when digital communication becomes more in tune with human interaction, a new wave of digital conversation is expected to emerge. Therefore, with increasing human interaction, users will likely feel a growing need for trust and comfort when using AI-driven solutions, fostering a stronger bond between people and technology.

This study offers detailed, comprehensive, and nuanced explanations of how peripheral signs convey human emotions. An in-depth inspection of every signal with its various permutations is achieved by utilizing more advanced machine learning techniques, such as RF and XGBoost. The study has exceptional accuracy in classifying various types of emotional states. The scale has increased the understanding to considerable lengths since it differentiates between good or bad feelings, as well as between active and passive. Most existing approaches analyze these signals in isolation, yielding suboptimal accuracies in predicting emotional states. Our study bridges this gap by employing a feature-level fusion of the signals to achieve a more holistic understanding of human emotions. Through rigorous pre-processing, feature extraction using the Fast Fourier Transform (FFT), and advanced classification using XGBoost, we achieved an accuracy of 90.57% in classifying the High Arousal, Low Valence (HALV) emotional state, surpassing the benchmarks set by previous studies. With emotional intelligence incorporated into the technology, the system can adapt to certain needs and preferences. It can be as simple as helping people with daily chores, making it easier for them to access these technological tools, or addressing mental health concerns. The approach could potentially alter how humans interact with machines. Technology must be targeted to not only understand emotions but also work with that knowledge to benefit people’s lives.

RELATED WORK

Current studies on emotion recognition systems explored diverse approaches that seek to improve the two major objectives of accuracy and real-time applicability. The report is about the crucial role of artifact removal in improving the quality of emotion prediction systems. The research focuses on eliminating environmental and experimental artifacts that can destroy signal quality. The suggestion is that even though artifact removal methods are necessary for accurate monitoring, particularly in personal care scenarios, developers should consider the computational cost and sensor quality to ensure real-time performance in home settings [1]. The view is consistent with the findings, which suggest that signal quality plays a crucial role in classification accuracy, especially in the context of emotion recognition based on physiological signals. The proposed a real-time framework to minimize generalization errors in stress recognition systems. Their approach harnessed non-linear features and bridged this model adjustment problem concerning the non-stationarity of signals typically achieved in practical applications. The work points out the capability of databases to validate the efficiency and accuracy of time-based classifications in emotion prediction systems, an aspect of research that has been challenging because of generalization problems [2].

The framework will significantly improve stress detection across various settings and provide a view of how advanced algorithms in ML can minimize classification errors. The authors examine subject-independent training methods, which have been widely accepted as the leading method to enhance the accuracy of models in emotion recognition. However, it is noted that these models have critical flaws when exposed to non-stationary signals in real-time analysis. Their research encourages more adaptive systems to handle variability in physiological signals that often arise in uncontrolled environments [3]. The contribution further enriches the ongoing discussion on improving emotion recognition systems, especially their reliability and robustness in dynamic environments. The topic is relevant to the study, which simplifies the complexity of emotional stimuli with a revolutionary solution like Virtual Reality (VR), which demonstrates that scenes in VR can provide a more appropriate induction method for arousal responses, making an immersive and controlled environment for tasks involved in emotion recognition. When applied with non-linear classifiers, the accuracy of the prediction model increases substantially according to the results. The challenge that arises is inducing the proper emotions within the participants, something often encountered when using traditional videos as media for evoking emotions [4]. The investigation in developing a semi-automatic stimuli selection method by using affective tags validated by ratings of participants. They examined EEG, Peripheral signals, and multimedia content analysis (MCA) in the classification of three binary classes of arousal level/valence and liking. Since this approach considers that each feature of the Gaussian Naive Bayes classifier is independent, it applies this classifier for classification. It was found that single-trial classification was better, as mentioned in [5]. The study on emotion classification was conducted by integrating features drawn from various physiological signals, and it was discovered

09 October 2026 04:16:41

that the use of Intrinsic Mode Functions (IMF)-based features markedly enhanced the classification rate. They utilized the AuBT dataset for their emotion classification endeavors. Additionally, the decision fusion method was adopted, which demonstrated superior performance compared to feature fusion. The study's findings underscored the efficacy of feature selection and fusion techniques in attaining precise emotion classification [6]. A real-time emotion recognition framework within human-computer interaction uses wearable devices like the body media sense, such as an armband and chest strap. In their study, they applied k-Nearest Neighbor (KNN) and Discriminant Function Analysis (DFA), obtaining coefficients from covariance matrices for the data matrix. A total input with emotive data was fed into discriminant functions, where a new dataset was categorized and sorted according to emotions. Conclusive results indicated that DFA succeeded best on datasets for fear, sadness, and anger, while KNN dominated frustration and surprise [7].

A multi-dimensional strategy using random matrix theory on Heart Rate (HR), GSR, and oxygen saturation (OXY) before feature extraction. They were thus able to characterize fluctuation structures using Local Scaling Dimension (LSD) power and improve the accuracy in predicting results [8]. A multi-resolution analysis of physiological signals using the AuBT dataset. They combined feature vectors from each emotion and found that EMG and RSP signals significantly influence emotion identification [9]. The attention was drawn to the importance of participant diversity in constructing subject-independent emotion-recognizing systems. It was observed that cross-validation improved the classifier's accuracy by more than 77.5% [10]. An emotion classification system intended to determine the emotional state of a user before his or her reaction. Their work achieved a significant accuracy of 95% in subject-dependent classification and achieved an accuracy of 70% in subject-independent classification through Emotion-specific Multilevel Dichotomous Classification (EMDC) [11]. The cross-validation of mean values of physiological signals sometimes resulted in a higher misclassification rate than classification rates [12]. A Multiple-Fusion-Layer based Ensemble Classifier of Stacked Autoencoder (MESAE) for emotion recognition to explore deep structures. In their developed model, Stacked Autoencoder (SAE) had three layers with some intricacy for removing the noise and stabilizing the feature representations. They combined the abstractions of the derived SAEs and processed them through a three-layer, adjacent graph-based network for the subsequent feature fusion and classification. The six sets of encodings they used were intricately fused through an adjacent graph-based network to enhance the aspect of feature fusion. The model has yielded an improved classification rate and F-scores of 5.2% by showing that MESAE outperforms the shallow and deep emotion classifiers for any instance size of physiological samples [13].

The minimal continuous dimensions for the classification of emotion. They used a multimodal fusion of both extensible and minimal data. The types of machine learning models they used were Support Vector Machine (SVM), Multi-Layer Preceptor (MLP), KNN, and classifiers. Among all the classifiers, SVM was successful and obtained an accuracy of 85.46% in sad emotion [14]. DEAP dataset was used, and it found that the accuracy of their model was highly influenced by the feature extraction phase rather than the classification method applied by [15]. Meanwhile, EEG signals and found that the dimensions of the first hidden layer, combined with a low learning rate, contributed significantly to achieving high classification accuracy [16]. Respiratory patterns focus on relationships between minimum airflow and volume, mean airflow and volume, and variance airflow and volume. They found that considering the total relationship between emotion and mathematical parameters improved model performance. The Area Under the Curve (AUC) and Recall readings for logistic regression indicated model improvement [17]. Compared unimodal, bimodal, and multimodal fusion techniques and found that multimodal fusion outperformed both bimodal and unimodal methods with an impressive accuracy of 72.2% for arousal and 78.4% for valence. Their study found that sweating caused significant oscillations in the signal captured, and the mean GSR value was related to the degree of arousal. Moreover, the results supported that multimodal fusion outperforms bimodal and unimodal approaches [18]. The use of multiple sensors significantly contributes to generalization and, therefore, brings about the improvement of accuracy in emotion prediction as it applies to different individuals. They reported that using more than one sensor can improve system performance for failure management [19].

Frequency domain features, along with voting criteria, are effective in reducing gender discrepancies in emotional responses. It was found that males have relatively higher arousal levels than females, while females are more fearful [20]. The artifact removal techniques and the techniques play a very important role, especially in real-time analysis. The techniques improve accurate monitoring and possibly enhance quality of life mainly in home settings and personal care scenarios. It emphasized the critical importance of tackling environmental, experimental, and systematic sources of artifacts while also underscoring the necessity of taking computational cost and sensor quality into account when creating real-time applications [21]. A real-time framework aimed at minimizing generalization errors in the recognition of stress. The used non-linear features for real-time applications and suggested that the dataset might be useful in confirming the effectiveness of real-time classification [22]. Most studies utilized subject-independent training methods, as these approaches typically resulted in higher accuracy. They also highlighted that this technique usually faces problems in practical implementations and further noted that the traditional analysis method struggles

09 October 2026 04:16:41

with non-stationary signals in real-time evaluations [23]. The problems of emotional stimulus videos are identified and proposed by pushing VR scenes as a powerful stimulus mechanism that could elicit responses by evoking emotions. Their experiment demonstrated the approach by enhancing prediction and demonstrating that non-linear classifiers could be used for real-time analysis [24].

MATERIALS AND METHODS

The Database for Emotion Analysis using Physiological Signals (DEAP) dataset is a widely utilized resource in emotion analysis studies. It comprises various physiological measurements, including EEG, EOG, EMG, and peripheral indicators, such as GSR, SKT, and PPG. The dataset comprises physiological recordings from 32 individuals and frontal facial video footage of 22 participants. The recorded emotional responses were evaluated using scales for arousal, valence, dominance, and liking and familiarity with the video. The Emotion analysis dataset was designed to identify human affective states through EOG, EMG, GSR, SKT, RES, and PPG data. The dataset contains results of spontaneous emotional trigger reactions to 40 one-minute music videos. It features data from 32 healthy participants, each exposed to 40 video stimuli through 40 experimental trials. The collected data includes signals from the central and peripheral nervous systems, providing a comprehensive view of physiological responses to emotional stimuli. The dataset comprises 32 participant responses and 40 videos, chosen from online annotated content and on their rating scales. Emotional responses were rated on arousal, valence, dominance, liking (on a continuous scale of 1-9), and familiarity (on a discrete scale of 1-5). Emotion intensity ranges from low (0 Hz) to high (9 Hz). The data collected includes 32-channel EEG signals at 512 Hz, peripheral physiological measurements, and facial recording for 22 participants. The observed bodily responses are used to represent emotions and can be used to enhance HCI by aligning with the user's emotional states. Emotions can be categorized in several ways, including the six basic emotions model using tree structures, dimension scales of emotion, and the valence-arousal scale proposed by Russell. The valence arousal model plots emotions on a Two-dimensional (2D) plane, with arousal and valence as in horizontal and vertical axes. The approach allows us to classify emotions as High Arousal and High Valence (HAHV), High Arousal and Low Valence (HALV), Low Arousal High Valence (LAHV), and Low Arousal and Low Valence (LALV), as portrayed in Figure 1.

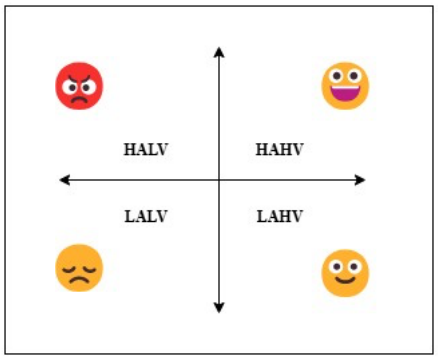


Figure 1. Russel's model with Arousal and Valence Scale

Arousal refers to the degree of activation or excitement of an individual experiences in response to a stimulus. It is evaluated using a continuum scale. Valence is also measured by continuum; it represents the positive or negative state of the experienced emotion. It is used to show the nature of the emotion, indicating if the feeling is pleasant or unpleasant. The valence-arousal model can be divided into four distinct quadrants: HAHV, HALV, LAHV, and LALV. Stimuli that induce emotional responses falling, but not any of these four quadrants, can be utilized to identify specific target emotions. The approach provides an understanding and categorization of emotional states based on their intensity of arousal and valence.

The proposed workflow for emotion recognition integrates peripheral signals, including SKT, RSP, GSR, EOG, EMG, and PPG, to achieve high accuracy in identifying emotional states. The data underwent preprocessing to remove noise and artifacts, ensuring clean signals for analysis. Origin software was used for signal extraction, smoothing, and transformation into a desirable format. The Savotikey method smoothens data points that fit the subsequent adjacent data points and converts to low degree polynomial. It works by sliding by sliding a fixed size window over the data and the specific feature signal with four window lengths and later performs filtering operations to extract frequency components of the signal frequency bands; the filtering operations used were Infinite Impulse Filter (IIR)Filter. Each

09 October 2026 04:16:41

signal was normalized within its specific range to minimize inter-subject and intra-session variability. Signal segmentation was performed to divide the data into fixed-length windows, enabling consistent feature extraction and analysis. Feature extraction was conducted using the FFT, transforming the time-domain signals into the frequency domain to capture significant patterns. Key features, including power spectral densities across frequency bands, along with statistical measures such as mean, standard deviation, and entropy, were derived for each modality. To further enhance model performance, features from all six signals were combined into a single feature vector, representing the unified physiological response of the subject. Redundant and less informative features were eliminated using feature selection techniques, streamlining the model's input and improving its efficiency. Figure 2 represents the entire workflow of the proposed model, which includes database, preprocessing, feature extraction, and model building.

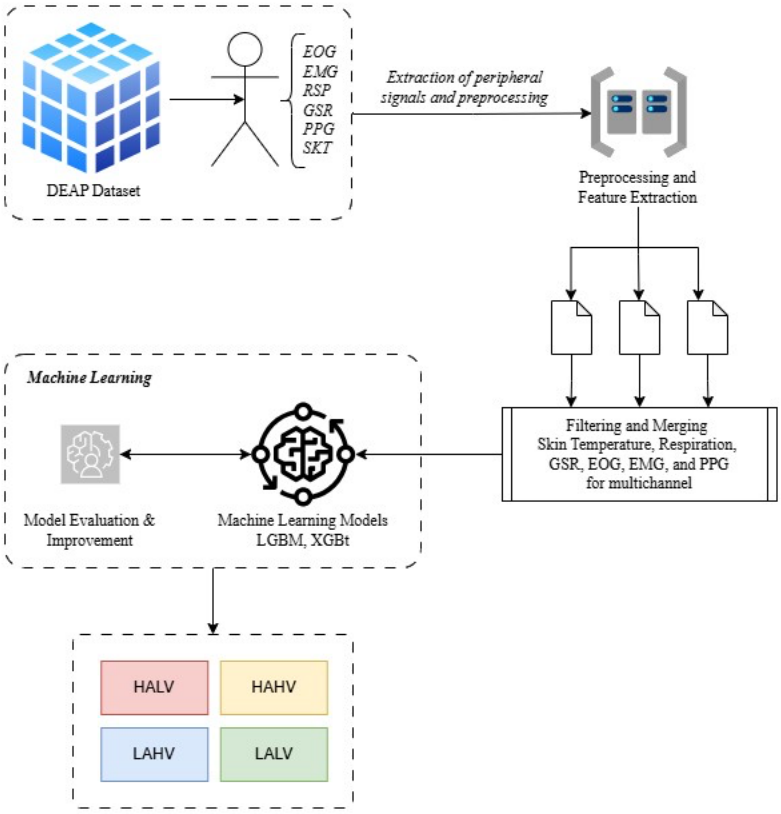


Figure 2. Proposed Workflow for Emotional Classification

Galvanic Skin Response (GSR), also known as electrodermal activity (EDA), is a continuous measurement of electrical parameters of the human skin, which depends on the variation of sweat reaction, which reflects the changes in the sympathetic nervous system. It measures the skin's resistance by placing two electrodes on the distal phalanges of the middle and index fingers. The mean value of GSR is highly related to the participant's arousal response level. The resistance is affected by the increase in perspiration caused by stress or surprise. Emotional changes cause sweat reactions that lead to changes in salt concentration in human skin, resulting in electrical resistance of the skin. When sweat glands become active, the moisture of the skin surface varies due to secretion and affects the electrical current flow property on the skin.

Photoplethysmography (PPG) measures the blood volume in the participant's thumb, which can also be used to compute the heart rate by identifying the local maxima; it is used to measure variations in heartbeats over time by analyzing the difference in the volume of capillaries such as interbeat periods and heart rate variability. The pleasant nature of a stimuli can increase the peak heart rate response It analyzes differences in each heartbeat cycle that are caused by the sympathetic and parasympathetic nervous system. Blood pressure and heart rate variability correlate with emotions correlates with emotions where stress can increase blood pressure. PPG measures blood volume changes using light sensors. PPG consists of a detector and red/infrared light-emitting diodes used as light sources for light sources to monitor tissue light intensity variations by transmission or reflection. Later, an FFT filter was applied

09 October 2026 04:16:41

to the filtered data to obtain signal information. The FFT, was applied to compute the Discrete Fourier Transform (DFT) of a signal. The pre-processed signal was then applied with IIR filter, Butterworth Filter with the frequency band of (0-0.5) Hz along with a band pass filter using stop frequency1(0.02), pass frequency1 of (0.05), pass frequency (0.5) and a stop frequency of (1) was applied. We then applied FFT filter on the filtered data to obtain signal information. The Fast Fourier Transform (FFT) was applied to compute the Discrete Fourier Transform (DFT) of a signal in Figure 3.

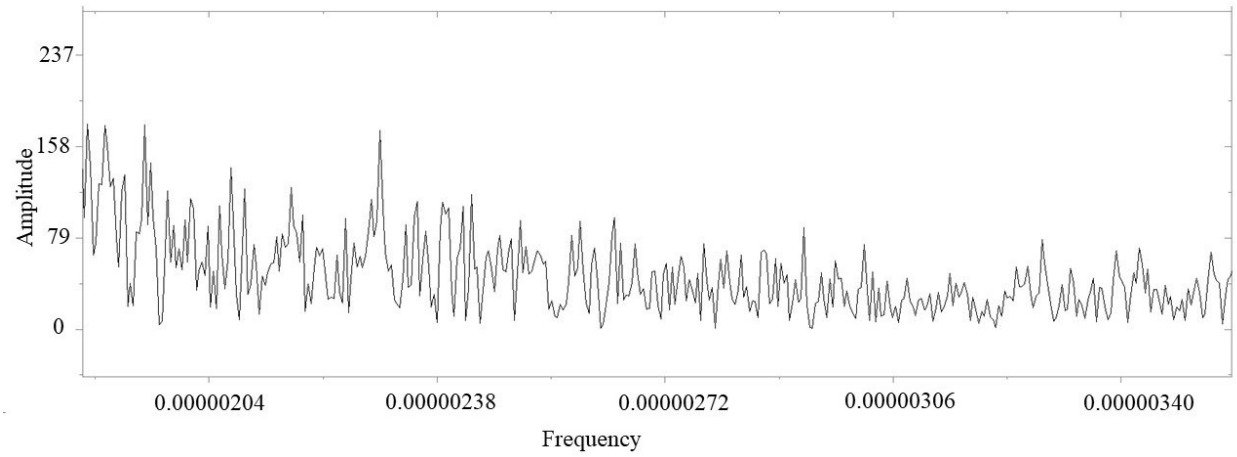


Figure 3. FFT on Filtered Plethysmograph

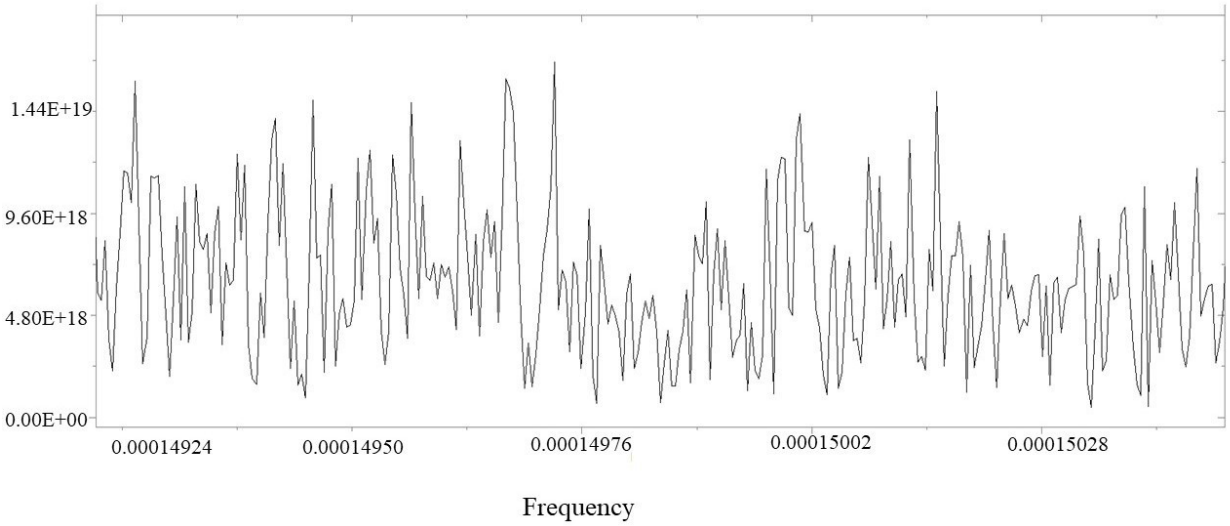


Figure 5. Smoothening of Horizontal Electrooculography

Respiration monitoring for emotion recognition offers valuable insights into emotional states by varying breath depth and velocity, reflecting different emotions. Our breathing patterns reflect emotions by changes in the breathing patterns. Shallower or faster breathing is closely related to emotions such as happiness, anger, or sadness, whereas slower breathing can indicate an emotional pace of calm or contentment. Heart rate can vary due to inspiration and expiration. Slow respiration is highly linked to relaxation, while irregular waveform or quick variations can indicate arousal emotions such as anger or fear. Similarly, it tends to suppress breaths when experiencing disgust. Some techniques for measuring respiration are humidity evaluation, temperature fluctuation measurement, oxygen concentration analysis, and monitoring body movements or respiratory sounds. Analyzing respiration for emotion recognition includes extracting meaningful features from breathing patterns, such as the volume of air inhaled and exhaled with each breath, Breath Patterns that examine the regularity or irregularity in breathing intervals, Respiratory Variability, which analyze the changes in the rhythm and variability of breathing over a period, to infer emotional states accurately. The signal is preprocessed by smoothening the Savitzky-Golay filter. The pre-processed signal was

09 October 2026 04:16:41

then applied with an IIR filter. Butterworth Filter with the frequency band of (0-2.4) Hz along with a band pass filter using stop frequency (0.02), pass frequency of (0.05), pass frequency (2.4), and a stop frequency of (3) was applied. Later, the FFT filters on the filtered data to obtain signal information as shown in Figure 4. The Fast Fourier Transform (FFT) was applied to compute the DFT of a signal.

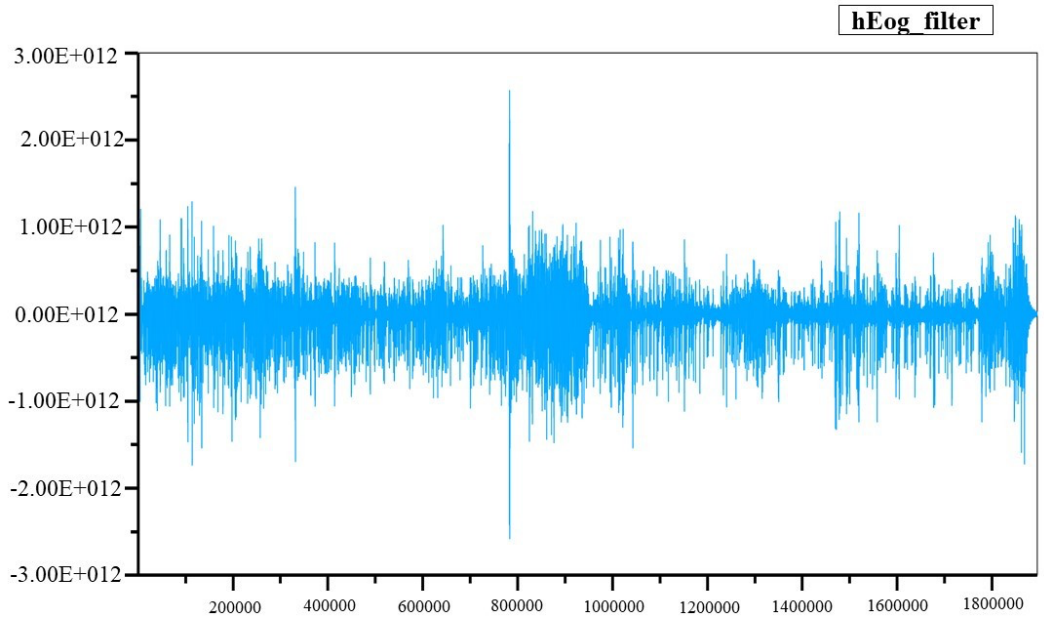


Figure 6. FFT on Horizontal Electrooculography

Skin temperature (SKT) serves as a valuable indicator for automatic emotion recognition due to its association with autonomic nervous system reactions. Controlled by sympathetic fibres, skin temperature decrease indicates emotional stress, anxiety, anger, or embarrassment. The studies reveal a correlation between emotional states and finger temperature changes, providing SKT's potential as an emotion indicator. Body temperature is handled by blood flow hence the process can trigger emotions which in turn trigger blood flow patterns that can be used to detect emotional fluctuations. Some measurement methods are contact sensors or non-contact thermal imaging. The accuracy of SKT can also be affected by where the sensor is placed in the user body due to slightly different blood flow patterns in the body. The preprocessing of the signal by smoothening of Savitzky-Golay filter. The pre-processed signal was then applied with IIR filter, Butterworth Filter with the frequency band of (0-0.2) Hz along with a band pass filter using stop frequency1(0.02), pass frequency1 of (0.05), pass frequency2(0.2) and a stop frequency of (1) was applied. Butterworth filters are characterized by their flat frequency response in the passband, ideally to maintain a constant level. The filter's behavior is overviewed by a transfer function, which is determined by the placement of its poles and zeros in the complex plane. The filter has uniformly distributed poles around the unit circle in this plane. The design ensures a smooth and consistent response. The filter effectively passes signals within a range while minimally reducing the amplitude. A notable feature of Butterworth filters is their sharp roll-off in the stopband, efficiently reducing the strength of the frequencies outside the passband. Then an FFT filter on the filtered data to obtain signal information, as shown in Figure 6: The FFT was applied to compute the DFT of a signal.

Electromyography (EMG) measures muscle activity by detecting surface voltages introduced by the phenomenon of muscle contraction; it records the electrical potential generated by muscle cells. The number of muscle fibers recruited in any given contraction depends on the force required for movement; the intensity of the electrical signal is proportional to the strength of the contraction. The EMG involves using electrodes to record the electrical activity of muscles and nerves that control muscle movement. The EMG signal consists of the activity of 2 facial muscles, Zygomaticus (zEMG) major and Trapezius(tEMG) major. The Trapezius major monitors muscle activity and records neck movements to investigate possible head movements when listening to music. The Zygomaticus major monitors the muscle activity and records the participant's laughs or smiles in response to a stimulus. The muscle activity features are obtained from the energy of EMG signals, with a frequency range of 4-40 Hz for different muscle activities. Electrooculography (EOG) measures the corneo-retinal resting potential between the front and back of the human eye. EOG signal measures the eye blinking activity and results in easily detectable peaks in the signal. Measuring eye movement involves using a pair of electrodes placed either above or below the eye or to the left or right of the eye. If

09 October 2026 04:16:41

the eye moves from its central position towards the electrodes, an electrical potential corresponding to the eye position is introduced between them. EOG is based on the principle that the front and back of the eye generate a potential difference as the eye moves. The eye blinking rate is highly correlated with the emotion of anxiety, fear, stress, or surprise. The smoothening applied on the EOG signals to easily able to remove the sharp peaks appears on the signals is shown in Figure 5. IIR filters are known for using feedback. The filter calculates results using not just current and past but also previous outputs, creating a loop where future outcomes depend on current outputs. The feedback mechanism allows the filter to retain information about past inputs, enabling it to achieve desirable frequency responses with fewer coefficients than FIR filters. In our work, we used an IIR filter and configured it with a bandpass filter based on the Amplitude of each signal; and implemented it using a Butterworth filter. The advantage of an IIR filter is its ability to ensure less memory usage, fast processing, and enable complex filtering tasks. We then applied an FFT filter on the filtered data to obtain signal information which is depicted in Figure 6. components. The FFT, was applied to compute the DFT of a signal.

The bandpass filter was used to obtain the band power for the signal features by using the power spectral density frequencies described by [5]in their work. The frequency bands for Respiration (0-2.4), Plethysmography (0.1–0.2 Hz), (0.2–0.3 Hz), (0.3–0.4 Hz), (0.01–0.08 Hz), (0.08–0.15 Hz), and (0.15–0.5 Hz) and Temperature (0–0.1 Hz) and (0.1–0.2 Hz). The Fourier Transform is a mathematical technique that breaks down a signal into its key component's frequencies, exposing its frequency elements. The Welch method was employed to extract the key signal features. The approach normalizes the Time Interval Square Amplitude (TISA). It derives various signal characteristics, including Frequency, Complex, Real, and Imaginary Components, as well as Magnitude, Amplitude, Phase, and Power in both TISA and decibel(dB) forms. The model worked on the amplitude and frequency of the results obtained for classifying emotions. The data was further split into train and test data and standardized, and 5-fold Stratified K-Fold cross-validation was done. We used Smote to balance the classes for better classification of emotions. We tried analyzing the results of the features individually and combined all the features to create a multimodal recognition system for emotion recognition. The algorithms used for analyzing the classification of emotions are RF, LGBM, and XGBoost.

RESULT AND DISCUSSION

The physiological signals used for classifying emotions were EOG, EMG, GSR, SKT, RES and PPG; the classification was based on Russell's arousal and valence scale. Smote was used to balance the model and enhance its performance. RF, LGBM, and XGBoost showed results for the classification. The top results are XGBoost, whose performance was outstanding compared with other models, as shown below in Table 1. The HALV emotional state is characterized by heightened arousal and negative valence, making it crucial for applications such as stress and mental health monitoring. XGBoost demonstrated its superiority by achieving an accuracy of 90.57%, significantly surpassing RF (89.94%) and LGBM (88.29%). The HALV state is often the most challenging to classify due to its subtle physiological cues. Yet, the fusion of features extracted from six modalities provided the depth and complementarity necessary for high precision. For the HAHV (High Arousal, High Valence) emotional state, which corresponds to positive emotions with heightened activity (e.g., excitement), XGBoost achieved an accuracy of 87.54%, outperforming RF (83.5%) and LGBM (85.52%). Similarly, in the LAHV emotional state, XGBoost attained the highest accuracy of 90.2%, as compared to RF (89.54%) and LGBM (85.62%). These results demonstrate XGBoost's ability to capture the nuanced differences between arousal and valence levels, making it a reliable model for classifying positive emotional states. The LALV emotional state, often associated with emotions like sadness or relaxation, was accurately classified by XGBoost with a performance of 88.6%. While RF and LGBM recorded lower accuracies of 83.71% and 82.08%, respectively, XGBoost's robust handling of high-dimensional, fused features contributed to its superior classification results in this category.

For classification, XGBoost was employed due to its capability to handle high-dimensional data and its robustness against overfitting. Hyperparameter optimization using grid search ensured the model's optimal configuration, maximizing classification accuracy. The model's performance was evaluated using metrics such as precision and recall rate of the Macro Average, which are detailed in Table 2. Cross-validation was performed to ensure that the model generalized well across different subsets of the data. This rigorous methodology enabled the identification of the HALV emotional state with an accuracy of 90.57%, surpassing the results of existing studies. XGBoost consistently outperformed both RF and LGBM across all emotional states, demonstrating its capacity to model complex interactions between fused features. Unlike Random Forest, which relies on decision trees without boosting, or LGBM, which uses a gradient-boosting framework optimized for speed, XGBoost combines scalability with advanced regularization techniques, making it particularly suited for handling the rich feature set derived from our multi-modal signals.

09 October 2026 04:16:41

Key Insights from the Results are summarized into three points: 1. Feature Fusion Advantage: Integrating features from six peripheral signals—ST, RSP, GSR, EOG, EMG, and PPG—played a pivotal role in achieving superior classification accuracy. By combining complementary information, the model could better represent the physiological responses associated with different emotional states. 2. Model Superiority: XGBoost not only handled the high dimensionality of the fused feature set but also leveraged feature importance to focus on the most relevant inputs, such as high-frequency components of EMG and mid-frequency components of RSP and GSR, which were crucial for distinguishing between arousal levels. 3. Comparative Performance: Across all emotional states, Random Forest demonstrated moderate performance, likely due to its lack of boosting capability, which is essential for fine-tuning complex datasets. LGBM showed competitive results but fell short due to its tendency to overfit some emotional states.

Table 1. Accuracy level of fusion features of eight peripheral signals with emotional states

Models/Emotional State	HAHV	HALV	LAHV	LALV
RF	83.50	89.94	89.54	83.71
LGBM	85.52	88.29	85.62	82.08
XGBoost	87.54	90.57	90.20	88.60

Table 2. Precision and recall rate of emotional classes

	Emotional states with Precision Rate				Emotional states with Recall Rate			
Models	**HAHV**	**HALV**	**LAHV**	**LALV**	**HAHV**	**HALV**	**LAHV**	**LALV**
RF	0.83	0.88	0.89	0.88	0.84	0.87	0.89	0.89
LGBM	0.87	0.87	0.85	0.82	0.87	0.88	0.86	0.82
XGBoost	0.86	0.9	0.9	0.87	0.86	0.9	0.89	0.88

The high accuracy rates achieved by XGBoost and the comparative performance with other ML algorithms depicted in Figure 7, particularly for the HALV state, have significant implications for real-world applications. Accurate identification of emotional states is critical for healthcare systems, where timely detection of negative emotional states like HALV can aid in mental health monitoring and suicide prevention. The robustness of the feature fusion approach and XGBoost's performance highlights the potential for deploying such systems in wearable devices and remote health monitoring setups, where multi-modal data streams can be processed in real time for early intervention.

09 October 2026 04:16:41

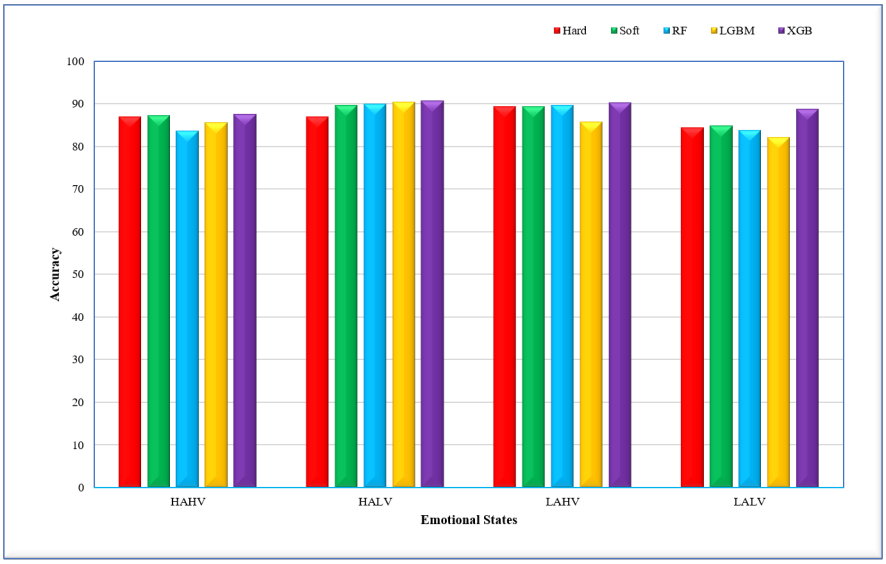


Figure 7. Evaluation Performance on ML models

By employing feature fusion and a sophisticated model like XGBoost, this study sets a new benchmark in emotion recognition from peripheral physiological signals. The results provide strong evidence that integrating multi-modal signals is not only feasible but also critical for achieving high accuracy in identifying emotional states. The ROC curve for our study, Figure 8, demonstrates a high AUC value of 0.90, indicating excellent discriminatory power. AUC values close to 1 signify that the model has a strong capability to distinguish between emotional states with minimal false positives. The steep rise of the curve near the top-left corner reflects the model's high sensitivity, ensuring accurate identification of positive cases. This performance, combined with the classification accuracies reported earlier, reaffirms the robustness of our approach to emotion recognition. The integration of feature fusion and the use of XGBoost not only enhanced accuracy but also provided consistent reliability across different emotional categories. The AUC score further validates the suitability of this methodology for real-world applications where precise emotion detection is critical, such as mental health monitoring and stress management systems. Emotion recognition using physiological signals has been a topic of significant interest in recent years, with several studies exploring using various signal modalities and ML classifiers. However, the reported accuracies in many of these studies have been limited, primarily due to individual signals or less advanced feature integration and classification techniques. The comparison of existing method with our study has been represented in Table 3. For instance, PPG and SKT signals with SVM classifier to classify basic emotions like happiness and sadness, achieving an accuracy of 76.48%. Whereas skin temperature, plethysmogram, skin conductance change, and respiration rate, but the combination resulted in a modest accuracy of 56% when classifying emotions into positive and negative categories. These studies demonstrate the challenge of achieving high accuracy when limited signals or simpler classification models are used. The respiration, skin conductance, EMG, and ECG signals with SVM to classify high-stress, low-stress, disappointment, and euphoria states, achieving an accuracy of 79.30%.

09 October 2026 04:16:41

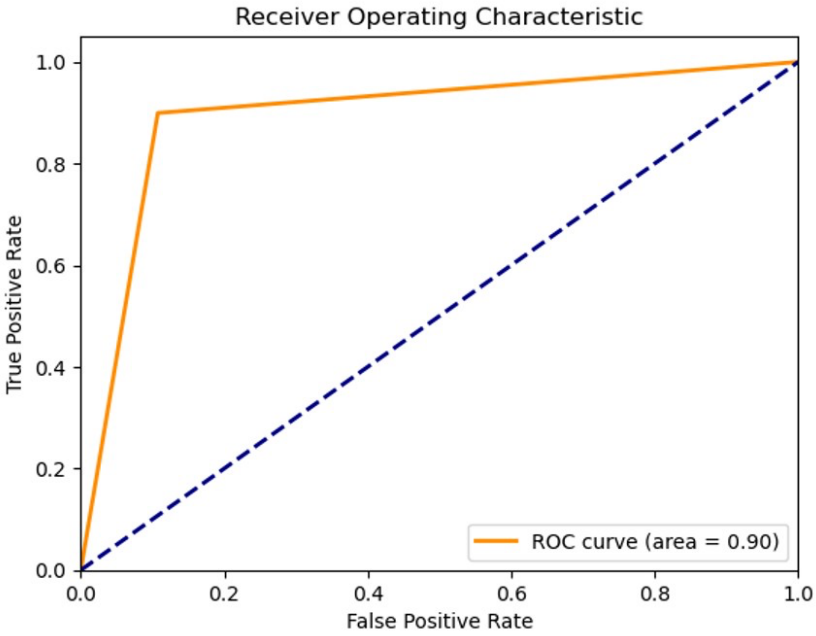


Figure 8. ROC curve of HALV

Table 3. Comparison of Emotion Recognition Studies

Ref	Modules Used	Classifiers	Accuracy	Emotions
[25]	Skin temperature, Plethysmogram, Skin Conductance, Respiration	SVM	0.56	Positive and Negative
[26]	GSR, RSP, EEG, ECG	LSTM	0.7236	Joy, Fun, Anger, Fear, Disgust, Neutrality
[27]	GSR, Blood Pressure, Chest Cavity	SVM	0.73	Calm, Neutral, Excited, Positive, Negative
[28]	PPG, SKT	SVM	0.7648	Happy, Sad
[29]	Respiration, Skin Conductance, EMG, ECG	SVM	0.793	High Stress, Low Stress, Disappointment, Euphoria
Our Study	Skin Temperature, Respiration, GSR, EOG, EMG, PPG	XGBoost	0.9057	HALV

Similarly, GSR, blood pressure, and chest cavity movements, attaining 73% accuracy for calm, neutral, excited, positive, and negative emotions. Although these studies utilized multiple modalities, their reliance on SVM limited the classification performance, particularly when dealing with complex emotional states. An alternative approach using deep learning was presented where GSR, respiration, EEG, and ECG signals were combined with a Long Short-Term Memory (LSTM) model.

09 October 2026 04:16:41

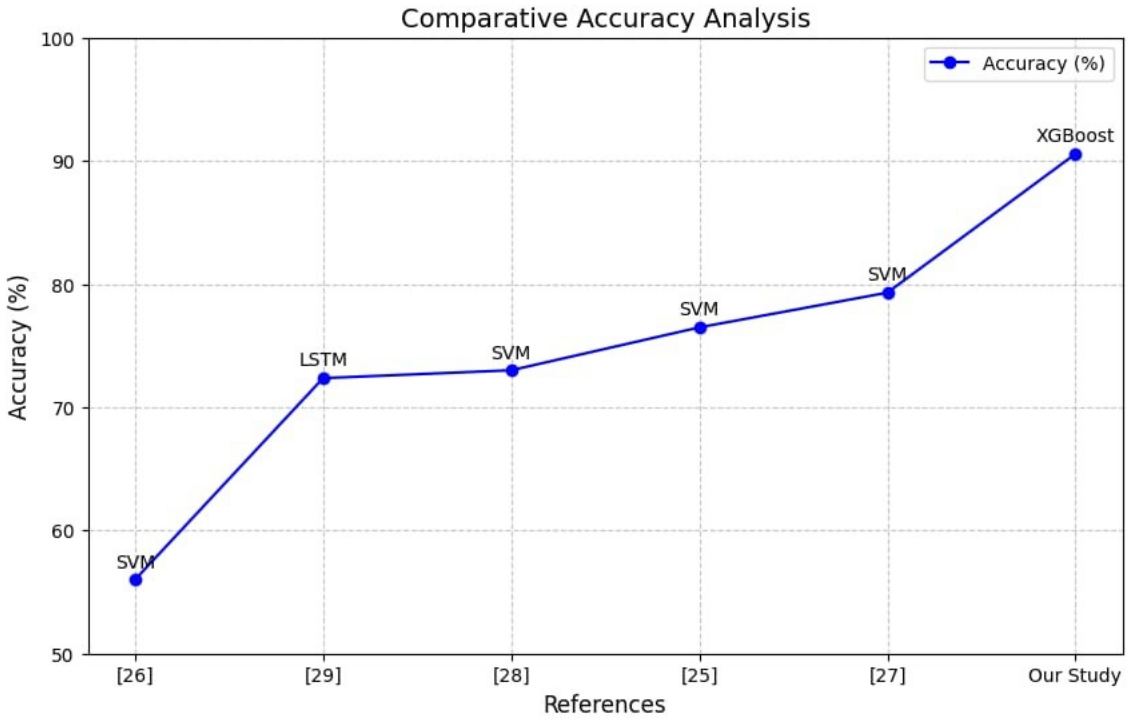


Figure 9. Performance evaluation of our model in comparison with existing works

This study achieved an accuracy of 72.36% for classifying emotions like joy, fun, anger, fear, and disgust. Despite leveraging deep learning, the lower accuracy highlights the challenges associated with handling multi-modal signals and the need for optimized feature fusion.

The Figure 9 shows an increasing trend in accuracy across studies, with [26] SVM having the lowest accuracy (56%) and Our Study XGBoost achieving the highest (90.57%). Studies using SVM or LSTM classifiers show moderate accuracy (72.36%–79.3%), while Our Study demonstrates the effectiveness of advanced features and the highest contribution given by peripheral signals like GSR with XGBoost. This highlights the importance of feature selection and advanced algorithms for superior performance in classification tasks. In comparison, our study employs a novel combination of six peripheral signals, Skin Temperature, Respiration, GSR, EOG, EMG, and PPG, and integrates them at the feature level. Using FFT for feature extraction and leveraging the advanced XGBoost classifier, our method achieved an accuracy of 90.57% for recognizing a broader set of emotions, including HALV. This represents a significant improvement over existing studies, especially in scenarios requiring the classification of complex emotional states.

CONCLUSION

The key objective here is to build a model that accurately detects and interprets human emotions through the skin surface. It is thought that this will result in a light, comfortable, and wearable device that can classify emotions well in HCI. The study indicates that with certain features, there can be different classifications of emotions, one of which is the valence-arousal scale proposed by Russell. This study demonstrates the effectiveness of feature fusion from peripheral physiological signals in accurately recognizing emotional states. By integrating six signals—SKT, RSP, GSR, EOG, EMG, and PPG and applying advanced feature extraction using FFT, we achieved a comprehensive representation of physiological responses associated with emotions. Utilizing XGBoost as the classification model, our approach attained an impressive accuracy of 90.57% for the HALV emotional state, outperforming both RF and LGBM. This accuracy rate is the highest reported for such a feature fusion approach and surpasses the results of existing studies that analyse peripheral signals individually or in less integrated forms.

The results underscore the importance of feature-level fusion in leveraging the complementary information from multiple modalities, which significantly enhances the predictive performance of emotion recognition systems. XGBoost's superior performance demonstrates its capability to handle high-dimensional data and identify key

09 October 2026 04:16:41

contributors to emotional state differentiation, such as EMG's high-frequency features and mid-frequency components from RSP and GSR. These findings hold significant implications for real-world applications, particularly in healthcare. The ability to accurately identify emotional states, especially negative ones such as HALV, can play a pivotal role in mental health monitoring, early detection of stress, and suicide prevention. The robustness and high accuracy of this approach make it suitable for integration into wearable devices and remote monitoring systems, enabling real-time emotion recognition and early intervention in critical situations. In conclusion, this study establishes feature fusion from peripheral signals and the use of XGBoost as a powerful framework for emotion recognition, setting a new benchmark in the field. Future work will focus on extending this methodology to real-time applications and expanding the emotional categories to include more significant nuanced states, further enhancing the system's utility in practical scenarios. The limitation of the study is that it is difficult to access the signal in real-time application.

REFERENCES

1. K. T. Sweeney, H. Ayaz, T. E. Ward, M. Izzetoglu, S. F. McLoone, and B. Onaral, "A methodology for validating artifact removal techniques for physiological signals," IEEE transactions on information technology in biomedicine 16, 918–926 (2012).
2. K. Tzevelekakis, Z. Stefanidi, and G. Margetis, "Real-time stress level feedback from raw ecg signals for personalised, context-aware applications using lightweight convolutional neural network architectures," Sensors 21, 7802 (2021).
3. Y. Liu and O. Sourina, "Real-time subject-dependent eeg-based emotion recognition algorithm," Transactions on Computational ScienceXXIII: Special Issue on Cyberworlds, 199–223 (2014).
4. R. Lima, A. Chirico, R. Varandas, H. Gamboa, A. Gaggioli, and S. B. i Badia, "Multimodal emotion classification using machine learning inimmersive and non-immersive virtual reality," Virtual Reality 28, 107 (2024).
5. S. Koelstra, C. Muhl, M. Soleymani, J.-S. Lee, A. Yazdani, T. Ebrahimi, T. Pun, A. Nijholt, and I. Patras, "Deap: A database for emotionanalysis; using physiological signals," IEEE transactions on affective computing 3, 18–31 (2011).
6. P. Gong, H. T. Ma, and Y. Wang, "Emotion recognition based on the multiple physiological signals," in 2016 IEEE International Conference on Real-time Computing and Robotics (RCAR) (IEEE, 2016) pp. 140–143.
7. F. Nasoz, C. L. Lisetti, K. Alvarez, and N. Finkelstein, "Emotion recognition from physiological signals for user modeling of affect," inProceedings of the 3rd Workshop on Affective and Attitude User Modelling (Pittsburgh, PA, USA (2003) pp. 1–8.
8. W. Wen, G. Liu, N. Cheng, J. Wei, P. Shangguan, and W. Huang, "Emotion recognition based on multi-variant correlation of physiologicalsignals," IEEE Transactions on Affective Computing 5, 126–140 (2014).
9. Z. Guendil, Z. Lachiri, C. Maaoui, and A. Pruski, "Emotion recognition from physiological signals using fusion of wavelet-based features,"in 2015 7th International Conference on Modelling, Identification and Control (ICMIC) (IEEE, 2015) pp. 1–6.
10. V. Kolodyazhniy, S. D. Kreibig, J. J. Gross, W. T. Roth, and F. H. Wilhelm, "An affective computing approach to physiological emotionspecificity: Toward subject-independent and stimulus-independent classification of film-induced emotions," Psychophysiology 48, 908–922 (2011).
11. J. Kim and E. Andre, "Emotion recognition based on physiological changes in music listening," IEEE transactions on pattern analysis and′ machine intelligence 30, 2067–2083 (2008).
12. S. Basu, N. Jana, A. Bag, M. Mahadevappa, J. Mukherjee, S. Kumar, and R. Guha, "Emotion recognition based on physiological signals usingvalence-arousal model," in 2015 Third International Conference on Image Information Processing (ICIIP) (IEEE, 2015) pp. 50–55.
13. M. K. Abadi, R. Subramanian, S. M. Kia, P. Avesani, I. Patras, and N. Sebe, "Decaf: Meg-based multimodal database for decoding affective physiological responses," IEEE Transactions on Affective Computing 6, 209–222 (2015).
14. V. Doma and M. Pirouz, "A comparative analysis of machine learning methods for emotion recognition using eeg and peripheral physiologicalsignals," Journal of Big Data 7, 18 (2020).
15. D. Ayata, Y. Yaslan, and M. E. Kamasak, "Emotion recognition from multimodal physiological signals for emotion aware healthcare systems," Journal of Medical and Biological Engineering 40, 149–157 (2020).
16. S. Tripathi, S. Acharya, R. Sharma, S. Mittal, and S. Bhattacharya, "Using deep and convolutional neural networks for accurate emotionclassification on deap data," in Proceedings of the AAAI Conference on Artificial Intelligence, Vol. 31 (2017) pp. 4746–4752.

09 October 2026 04:16:41

17. R. A. Hameed, M. K. Sabir, M. A. Fadhel, O. Al-Shamma, and L. Alzubaidi, "Human emotion classification based on respiration signal," inProceedings of the International Conference on Information and Communication Technology (2019) pp. 239–245.
18. Q. Zhu, G. Lu, and J. Yan, "Valence-arousal model based emotion recognition using eeg, peripheral physiological signals and facial expression," in Proceedings of the 4th International Conference on Machine Learning and Soft Computing (2020) pp. 81–85.
19. D. Ayata, Y. Yaslan, and M. E. Kamasak, "Emotion based music recommendation system using wearable physiological sensors," IEEE transactions on consumer electronics 64, 196–203 (2018).
20. H. U. R. Siddiqui, H. F. Shahzad, A. A. Saleem, A. B. Khan Khakwani, F. Rustam, E. Lee, I. Ashraf, and S. Dudley, "Respiration basednon-invasive approach for emotion recognition using impulse radio ultra wide band radar and machine learning," Sensors 21, 8336 (2021).
21. K. T. Sweeney, T. E. Ward, and S. F. McLoone, "Artifact removal in physiological signals—practices and possibilities," IEEE transactions oninformation technology in biomedicine 16, 488–500 (2012).
22. M. Li, L. Xie, and Z. Wang, "A transductive model-based stress recognition method using peripheral physiological signals," Sensors 19, 429 (2019).
23. H. U. R. Siddiqui, K. Zafar, A. A. Saleem, M. A. Raza, S. Dudley, F. Rustam, and I. Ashraf, "Emotion classification using temporal andspectral features from ir-uwb-based respiration data," Multimedia Tools and Applications 82, 18565–18583 (2023).
24. L. Shu, J. Xie, M. Yang, Z. Li, Z. Li, D. Liao, X. Xu, and X. Yang, "A review of emotion recognition using physiological signals," Sensors18, 2074 (2018).
25. K. Takahashi, S.-y. Namikawa, and M. Hashimoto, "Computational emotion recognition using multimodal physiological signals: Elicitedusing japanese kanji words," in 2012 35th International Conference on Telecommunications and Signal Processing (TSP) (IEEE, 2012) pp. 615–620.
26. T. Song, W. Zheng, C. Lu, Y. Zong, X. Zhang, and Z. Cui, "Mped: A multi-modal physiological emotion database for discrete emotion recognition," IEEE Access 7, 12177–12191 (2019).
27. G. Chanel, K. Ansari-Asl, and T. Pun, "Valence-arousal evaluation using physiological signals in an emotion recall paradigm," in 2007 IEEE International Conference on Systems, Man and Cybernetics (IEEE, 2007) pp. 2662–2667.
28. M. W. Park, C. J. Kim, M. Hwang, and E. C. Lee, "Individual emotion classification between happiness and sadness by analyzing photoplethysmography and skin temperature," in 2013 Fourth World Congress on Software Engineering (IEEE, 2013) pp. 190–194.
29. C. D. Katsis, N. Katertsidis, G. Ganiatsas, and D. I. Fotiadis, "Toward emotion recognition in car-racing drivers: A biosignal processing approach," IEEE Transactions on Systems, Man, and Cybernetics-Part A: Systems and Humans 38, 502–512 (2008).

09 October 2026 04:16:41